India Faces a Major Dilemma in Coal

India has the third largest reserves of coal and generates more than 70 per cent of its electricity at present by using coal. Coal was the cheapest, easily available and most reliable source of power generation for the country for decades. All other fossil fuels—gas, crude oil—were in short supply. It was a no-brainer that India went after constructing large coal-based thermal power plants to generate electricity for its power-starved industry, agriculture and households.

However, coal has created a major pollution problem in the country. While black soot coming out of coal-fired power plants was visible and therefore got a quicker policy response, the more dangerous emissions of nitrogen oxides, sodium oxides, lead and mercury, being invisible, were practically ignored by policymakers until very late in 2015. India has now established standards for their emission and the thermal power generation industry is in the process of installing requisite technology to capture these pollutants and control their emissions within permissible limits. The work is progressing very slowly.

In the past few years, renewable technology has advanced significantly, bringing down the cost of generating electricity to levels lower than coal power. Grid technology is also advancing to manage the volatility of renewable power. Battery technology is improving as well, which provides the option of storing renewable power. Advancement in hydrogen-based power generation is likely to provide good options for mainstreaming the volatility caused to the grid by renewables.

While the historical carbon contribution of India is low, today the country is the third largest emitter of carbon in the environment. India faces a big dilemma in persisting with coal power generation.